# Planar Keller Maps after the Counterexample: Machine-Certified Newton-Polygon Exclusions, the Staircase Transport, and the Recalibrated Frontier

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CAOS Research program, `github.com/fsantibanezleal/CAOS_RESEARCH`

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**Abstract**

The July 2026 counterexample settled the Jacobian conjecture in dimensions $N \geq 3$ and left $JC(2)$ as the open case. This paper develops an exact-arithmetic program on the planar case. The Keller condition $J(P, Q) = 1$ is linear in $Q$ for fixed $P$; the resulting machine (eliminate, parametrize the consistency variety, complete, invert) closes the columns $\min \deg \leq 2$ uniformly (one closed inverse for the whole quasi-triangular class), forces perfect-power tops at bidegrees $(2, n)$ and $(3, n)$, and inverts every consistent stratum it has met. Four unconditional exclusion theorems follow from a weight-class calculus: a two-term $P = x + ax^u y^v$ is never a Keller component at any partner degree; neither is any lower-weight perturbation of it; no Keller component carries a Newton top edge anchored at the linear vertex $x$ with interior direction; and if $x$ is a vertex of the Newton polygon then $P = x + f(y)$ exactly (the vertex dichotomy). The residual open core is the transport of the Keller constant along the staircase of Newton vertices swallowing $x$: we show the completion window is block-triangular under the edge grading, exhibit the classwise transport chain as an executable elimination whose obstruction always localizes at the constant's class, and prove the corresponding fifth exclusion: on the certified grid shapes with a primitive top the certificate-tower induction closes it at all partner degrees (cleared base certificates with monomial pairings such as $-13860\, a^7$, plus a resonance-reduction lemma showing window growth can never repair the constant's row when the top form is not a proper power). A recalibration against the primary sources shows that our composite-gcd certificates replicate literature-covered territory (the verified floor is gcd $\leq 8$, primes, $2p$, then Heitmann's $B \geq 16$ and the Guccione–Guccione–Valqui dichotomy $B = 16$ or $B > 20$), and that the open frontier is the $B = 16$ normal form together with the single surviving degree pair $(72, 108)$ below 125. For that open pair, an exact instantiation of the primary sources gives two necessary conditions on the original pair: seven possible northeastern vertices for a nonzero inner polynomial, and $\deg_x \operatorname{Res}_y(P, Q) = 21$ with an $84 + 24$ major/minor root partition; neither condition alone excludes the pair. A constructible determinantal-strata calculation closes the first actual three-coefficient GGHV

slice exactly. On the residual curve, two normalized maximal minors reduce, with $z = u^9$, to $21-96z-1024z^2$ and $(8z+1)^{14}$; an explicit integer Bezout identity has constant $17^{14}$. After adding $\varepsilon_{(0,7)}$, the only two-chart residual is a degree-12 fiber over $z = -1/8$. A third exact minor has coprime degree-13 fiber determinant, certified by a second integer Bezout identity. A connected mod-9 row/column grading extends to every nonconstant direction in the 26-parameter lower family. This is a strict slice result and does not exclude the full $(72, 108)$ family. At the level of the full family, after removal of the structural constant column and with the complete 302-row pool (the forced-only construction has 289 rows), one selected augmented minor compresses to a 36-column block depending on 24 of the 51 parameters. Two connectivity-generating triples factor exactly; one graph-active direction cancels identically, while the other determinant becomes weighted homogeneous with weights $(7, 3, 9)$ after shifting the forced coefficient. On the $d = 0$ boundary, the tautological kernel $[P, P] = 0$ can be quotiented exactly. One quotient determinant factors into $b^{32}$, one five-term weighted factor, and five binomials; a second is a nonzero scalar times $a^{107}$. Together with the exact origin rank gap, these charts close the complete quotient boundary. On $d \neq 0$, a first alternative chart reduces each selected component $G, L, Q$ to a finite intersection. A third exact determinant factors as $X^{30}LQR$, for $X = A^3$, and exact component ideals close $G$. A row basis selected independently on the finite $L$ and $Q$ residuals is in fact shared. Its 23-term determinant descends as $A^{87}R(A^3, B)$; exact unit-ideal certificates close both residuals. Together with the quotient boundary this closes the complete three-parameter $T_B$ restriction. This is still a strict slice result, not closure of the 24-parameter core, the full family, $(72, 108)$, or $JC(2)$. Restoring direction $(2, 9)$ gives the first higher-dimensional recursive atlas. The shared determinant is $A^{87}(R(X, B) + YS(X, B))$, $X = A^3$, $Y = A^2C$, with $\gcd(R, S) = 1$. A second exact minor covers a dense open of this rational graph; its residual factors into three plane curves $F_3F_6F_7 = 0$. On the linear-in-$X$ component $F_3 = (5B + 4)^3 + 16X$, a third exact minor reduces the principal-open residual to $Q_9(B)Q_{15}(B) = 0$: 24 normalized values, or 72 lifted algebraic points. On the quadratic-in-$X$ component $F_6$, the same exact section has a nonzero linear quotient class $U(B)X + V(B)$. Its degree-74 function-field norm has five irreducible factors of degrees $2, 3, 6, 18, 30$. Exact same-point arithmetic places the first three on $X = 0$ or $R = S = 0$; only the degree-18 and degree-30 factors survive on $AS \neq 0$. Thus $F_6$ also becomes finite: 48 normalized values, or 144 lifted algebraic points. A distinct exact section on $F_7$ has a degree-58 norm. Its $B^{16}$ factor lies on $A = 0$, and its degree-12 factor lies on $R = S = 0$; the surviving irreducible degrees $3, 9, 18$ give 30 normalized values and 90 lifts. Exact CRT comparison then identifies the degree-9 and degree-18 blocks with the same geometric points already projected from $F_3$ and $F_6$, reducing the seven factor appearances to five squarefree blocks of total degree 75. The $h_7$ section covers the unique degree-15 and degree-30 blocks. Two further exact maximal-minor sections generate the unit ideal on the remaining $F_7$ blocks of degrees $3, 9, 18$. Hence the accumulated atlas closes the entire $AS \neq 0$ rational graph. Exact factorwise subresultants then identify the principal-open base locus as a reduced 90-dimensional algebra with field-block degrees $3, 6, 12, 69$. Existing sections cover the degree-12 and degree-69 blocks; one new exact section completes degrees 3 and 6. Bezout identities, independent resultants, multiplication-matrix norms, and a direct determinant control certify the finite atlas uniformly in $Y$. Thus the complete $A \neq 0$ sector of the four-parameter restriction is covered. Returning to the original 302-by-125 augmented matrix before invariant division, the direct $A = 0$ boundary admits two exact minors independent of $C$. Their divisors are $(5B + 4)^3(25B^2 - 20B + 16)^3$ and $B^{95}(109375B^6 - 110592)(21875B^6 - 4800B^3 - 24576)$, up to nonzero scalars. An explicit Bezout identity makes their gcd one. Combined with the $d = 0$ quotient cover, this closes the complete four-coefficient restriction. Adjoining the transverse coefficient at $(2, 8)$, EXP-132 closes the entire direct $A = 0, d = 1$ boundary of that five-coefficient restriction. Three normalized maximal minors generate the unit ideal in $\mathbb{Q}[B, C, T]$: after one exact row combination, two depend only on $B$ and are coprime, with an explicit characteristic-zero Bezout certificate [MV/D]. The inherited EXP-123 and EXP-124 bases vanish identically on this

specialization, so the certificate requires a new SCC-guided basis rather than a parameter substitution into the four-coefficient atlas. EXP-135 then proves that the EXP-124 section is unchanged by $(2, 8)$ on its complete normalized $(A, B, C)$ ambient chart. The transverse core has ranks $7, 3, 0$ through its first three powers. Separate determinant degree bounds $(25, 24, 6, 7)$, a complete 136,500-control tensor grid over 30 primes, and a CRT modulus exceeding the explicit characteristic-zero coefficient bound certify $\det(H + TK_{(2,8)}) = \det(H)$ [MV/D]. Hence the $F_3F_6F_7$ divisor of the four-coefficient restriction is retained for this section. The other residual sections and the transverse $d = 0$ quotient remain open, so the complete five-coefficient restriction and every larger target remain open. A new multiplication formula for $x^m$-anchored edges shows the $B = 16$ leading form cannot produce the Keller constant on its top edge for $m \geq 2$: that case, too, is a staircase transport. Helali has claimed, with exact certificates, that both coefficient systems of GGHV Proposition 4.3 for $(72, 108)$ are empty. We verify this independently in part: our own derivation reproduces his equations and his degree-35 eliminant, our own computation proves the second configuration empty, all of his identities for the first configuration hold in our exact arithmetic, and the first configuration reduces to the unit ideal at five primes (Section 8). All computations are exact and reproducible; every claim is labeled machine-verified [MV], derived [D], or conjectural [C].

# 1 Introduction

Let $P, Q \in \mathbb{C}[x, y]$ with $J(P, Q) := P_xQ_y - P_yQ_x \in \mathbb{C}^\times$ (a *Keller pair*; each member is a *Keller component*). The two-variable Jacobian conjecture $JC(2)$ asserts every Keller pair is an automorphism pair. Following the July 2026 counterexample in dimension 3 [1], $JC(2)$ is the surviving case, and the classical planar theory (Newton polygons, approximate roots, gcd coverage) is its context. The three-dimensional counterexample itself (validation, the seed family, the geometry of non-properness, two-dimensional equivariant rigidity) is studied in the companion paper [22]; this paper treats the planar case. Every computer-assisted statement cites the archived computation that establishes it (EXP-*nnn*; see the section on data and code availability).

Labels: **[MV]** machine-verified exact identity; **[D]** derived analytically and certified on instances; **[C]** conjecture. Machine verification is exact arithmetic: sympy over $\mathbb{Q}$ or $\mathbb{Q}(i)$, and modular arithmetic over prime fields with Chinese-remainder reconstruction and explicit degree and height bounds where a statement over $\mathbb{Q}$ is derived from modular computations; no floating-point value enters a certificate.

Two structural inputs from the companion are used freely. First, every $\mathbb{G}_m$-equivariant Keller map of $\mathbb{C}^2$ is an automorphism; on the opposite-sign scope relevant to the three-dimensional mechanism it is linear, and with one zero weight it is affine (the corrected statement of the companion; independently classified for all signatures by Shaska [21]). Second, at every weight ray the leading pair of a planar Keller map is Jacobian-dependent (the tropical bridge, EXP-013), which is the entry point of all edge analyses below.

# 2 The planar machine

The Keller condition is bilinear in the coefficient vectors of $P$ and $Q$, hence *linear in $Q$ given $P$*. In the affine gauge (identity linear parts) this yields a machine per degree pair: eliminate $Q$ (the consistency ideal in $P$'s coefficients), parametrize, complete linearly, and invert or fiber-test (EXP-017, EXP-019, EXP-020).

**Theorem 2.1** ([MV]; uniform closure at $\min\deg \le 2$ (EXP-021/022))**.** *Write $\ell = \alpha x + \beta y$, $\beta \neq 0$. In shear coordinates the Keller condition for $P = x + f(\ell)$ (any polynomial $f$) is a linear first-order PDE with characteristic invariant $P$; its complete polynomial solution space is $Q = \ell/\beta + H(P)$, $H$ arbitrary, and the pair inverts by one closed formula. Consequently every planar Keller map with $\min(\deg P, \deg Q) \le 2$ is invertible, and the whole quasi-triangular class $P \sim x + f(\ell)$ is invertible at every degree.*

**Theorem 2.2** ([MV]; forced tops at $(2,n)$ and $(3,n)$ (EXP-019/020/022/039))**.** *The $(2,3)$ consistency ideal is the single equation $(4A_0A_2 - A_1^2)^2 = 0$ (P's top form is a perfect square); the $(2,4)$ elimination returns the same ideal. At $(3,4)$, the $6\times 5$ linear map $Q_4 \mapsto J(P_3, Q_4)$ is rank-deficient exactly when $P_3$ is a perfect cube: all $5\times 5$ minors vanish on the cube parametrization, every minor lies in the Hessian-covariant ideal, and completeness follows from $GL_2$-equivariance ($J(f\circ A, g\circ A) = \det(A)\, J(f,g)\circ A$, verified identically) plus full rank at the two non-cube orbit representatives $x^2y$ and $xy(x+y)$. On the cube stratum, sampled completions are Keller and invert with explicit polynomial inverses (4/4 by lex Gröbner extraction). At $(3,3)$ the consistency ideal forces the quasi-triangular alignment at the radical level.*

The degree columns close by classical coverage: $\gcd(2,n) \le 2$ and $\gcd(3,n) \in \{1,3\}$ are inside the verified gcd landscape of Section 5, so the machine's contribution there is the *constructive* loop (explicit consistency varieties and inverses), stated as replication. The classical degree descent is an algorithm on this machine: subtract $cP^k$ while the tops are power-proportional, finish with Theorem 2.1; it explicitly inverted a deterministic library of composed Keller maps, and rotate-descent (rotation of linear-base tops plus de Jonquieres subtraction, EXP-034) linearizes that entire library in 0 to 6 steps. Its exact limit: mixed-corner tops stay mixed under rotation; those are the classical hard shapes, and they are where the exclusion theorems below take over.

## 3 Four unconditional exclusions

Throughout, $P$ has linear part $x$ after affine gauge. The weight-class calculus: under $w = (v, 1-u)$, the operator $J(P,\cdot)$ shifts weight by a constant, so the Keller equation decouples into weight classes; the technique goes back to Dixmier and Joseph [17] and is standard in the polygon literature [8]; the statements below appear to be new in the stated generality (Section 6).

**Theorem 3.1** ([MV/D]; the weight-class theorem (EXP-029))**.** *For $u \ge 2$, $v \ge 1$, $a \neq 0$, the polynomial $P = x + a\,x^u y^v$ is not a Keller component: no $Q$, of any degree, satisfies $J(P,Q) \in \mathbb{C}^\times$.*

*Proof shape.* $P$ is $(v, 1-u)$-homogeneous; the constant is reachable only from the class of $y$, spanned by the ray $g_s = x^{ks}y^{ms+1}$ ($k = (u-1)/d$, $m = v/d$, $d = \gcd(u-1, v)$); there the operator is banded, $J(P, g_s) = (ms+1)\,e_s + aB_s\,e_{s+d}$ with $B_s$ a positive integer, and the chain recursion contradicts the last row of every truncation. □

**Theorem 3.2** ([MV/D]; lower-weight perturbations never rescue (EXP-030/031))**.** *With $u, v, a$ as above and $R$ any polynomial whose monomials have degree $\ge 2$ and $(v, 1-u)$-weight strictly below $v$: $P = x + a\,x^u y^v + R$ is not a Keller component.*

**Theorem 3.3** ([MV/D]; every $x$-anchored edge falls (EXP-032))**.** *Let $z = x^k y^m$, $k, m \ge 1$, and $E = x\,\varphi(z)$, $\deg\varphi = g \ge 1$. On the $y$-class the edge operator is* multiplication*: $J(E, g_s) = [(ms+1)\varphi + kz\varphi']z^s$, and the top coefficient $(mD + kg + 1)\varphi_g$ of the class ODE kills every*

*polynomial candidate. Hence $E$ is never the leading form of a Keller component, and $P = E + R$ (any strictly lower-weight tail) is never a Keller component.*

**Theorem 3.4** ([MV/D]; the vertex dichotomy (EXP-033))**.** *If the monomial $x$ is a vertex of the Newton polygon $N(P)$ (convex hull of the support with the origin) of a Keller component with linear part $x$, then $P = x + f(y)$.*

**Lemma 3.5** (Annihilation (EXP-036))**.** *For any polynomial $P$, $J(m, P^k) = -k\,L(P^{k-1}m)$ with $L = J(P, \cdot)$ (Leibniz). Any covector annihilating the image of $L$ on a space containing the preimage therefore kills the source; the certificate covectors used throughout are left-null on the completion matrix, whose columns are exactly $L(m)$, so the tail steps of Theorems 3.2–3.4 are unconditional. The finite-window subtlety (spurious pairings for out-of-window preimages) reproduces the computed values exactly.*

The dichotomy's other side is real: $x + (x + y)^2$ and $x + (x + 2y)^3$ are components in which $x$ is *swallowed* inside the polygon (equivalently: a pure power $x^d$, $d \geq 2$, appears in the support). Every non-triangular component is of that kind. The residual open core is therefore precise: components whose polygon swallows $x$ with a mixed staircase.

## 4 The corner is diagonal; the staircase transport

**Proposition 4.1** ([MV]; the diagonal corner (EXP-035))**.** *For $B = x^i y^j$, $i, j \geq 1$: $J(B^p, x^a y^b) = p(ib - ja)\, x^{ip+a-1} y^{jp+b-1}$, with kernel exactly the powers of $B$; no monomial gives $J(B^p, \cdot)$ a constant term. The mixed corner is clean; the obstruction lives on the staircase of Newton vertices joining the swallowed linear vertex to the corner.*

**Theorem 4.2** ([MV]; block triangularity and the transport chain (EXP-037))**.** *Fix the $x$-edge grading $w = (v, 1 - u)$ and let $s_0$ be the minimal $w$-class of $P$. In the completion window (unknown $Q$ of bounded degree), every nonzero matrix entry sits at a class offset $s - \sigma$ for a $P$-class $s$ ($\sigma = v + 1 - u$), and the offset $s_0 - \sigma$ is minimal: ordering $Q$-classes by $w$-class, the system is block triangular. Classwise elimination (solve the diagonal block, inject its kernel parameters, emit its cokernel obstructions) is therefore well-defined; it reproduces the outcome of the monolithic consistency test on every tested sample, including a consistent positive control. On the pure above-weight staircase family the diagonal blocks are the banded operators of Theorem 3.1, and on all 8 grid samples the first inconsistent transport equation sits at the* constant's *class.*

**Theorem 4.3** ([MV]; the fifth exclusion, window form (EXP-042))**.** *Let $P = x + a\,x^u y^v + b\,x^d$ ($u \geq 2$, $v, d \geq 1$; the minimal shape swallowing $x$). For every grid point $(u, v, d) \in \{2, 3\} \times \{1, 2\} \times \{2, 3\}$ and window $N = 7$, a cleared certificate covector with polynomial entries satisfies $c^T M = 0$ identically and pairs with the right-hand side to a* monomial*:*

$$-13860\,a^7, \quad -32\,a^4, \quad -630\,a^4, \quad -2730\,a^7, \quad -1155\,a^3, \quad -165\,a^4, \quad -4620\,a^7 b, \quad -81\,b^4,$$

*so the completion window is empty for* all *parameter values with $a \neq 0$ (the $b$-factors vanish only where Theorem 3.1 applies anyway). At $(u, v, d) = (2, 1, 2)$ the pairing at window $N$ is $-c_N\,a^N$ with $c_N \in \{420, 1260, 13860, 180180, \dots\}$ for $N = 5..9$: a nonzero monomial in $a$ alone at every certified window. The annihilation mechanism of Lemma 3.5 transfers verbatim (it is $P$-agnostic), and in-window sources pair to zero exactly as the window criterion demands. A three-parameter tail certificate ($-27720\,a^7 c_3^4$) extends the family.*

**Lemma 4.4** ([MV/D]; the tower lemma (EXP-044/046)). *Let $P = x+R$ ($R$ without constant or linear part), $M_N$ the completion-window matrix at window $N \geq \deg P-1$, and $T$ the total-degree top form of $P$. Suppose every form in $\ker J(T,\cdot)$ at each degree is a scalar multiple of a power that reduces, modulo $\mathbb{C}[P]$, to degree $\leq N$ (this holds whenever $T$ is not a proper power, in particular when $T$ is a primitive monomial, or $T = P - x$ with $P - x$ not a proper power, by expanding $T^k = \lambda P^k + \text{lower}$; it fails for proper powers, for example a pure power $T = bx^d$, whose kernel contains every $x^M$). Then any left-null certificate of $M_N$ with nonzero pairing extends to $M_{N+1}$ with the same (scaled) pairing: window inconsistency propagates to every larger window. Proof ingredients, each machine-verified: old columns vanish on new rows (degree bookkeeping); the new block is $J(T,\cdot)$ on degree-$(N+1)$ forms with the predicted kernel; each resonance $\kappa$ satisfies $\kappa - \lambda P^k \in$ window with $P^k \in \ker L$, so $L(\kappa)$ is an in-window image which every certificate annihilates; the extension was realized explicitly across an active resonance (window $8 \to 9$, ratio 2, pairing preserved). The measured window law ($-c_N a^N$ pairings with the odd-prime $2N-3$ ratios) is this tower seen through the clearing normalization.*

**Theorem 4.5** ([MV/D]; all-degree fifth exclusion (EXP-042/044/046)). *For $(u,v)$ one of $(2,1)$, $(3,1)$, $(3,2)$ and $d$ one of 2, 3 (the grid points of Theorem 4.3 with $x^u y^v$ primitive), and any $a \neq 0$, $b$: $P = x + a\,x^u y^v + b\,x^d$ is not a Keller component at* any *partner degree. (Base: the cleared certificates of Theorem 4.3; induction: Lemma 4.4, whose hypothesis holds because the top form is the primitive monomial $a\,x^u y^v$ or, at $(u,v,d) = (2,1,3)$, equals $P - x = x^2(ay+bx)$, which is not a proper power.) Outside the grid no base certificate is available: Theorem 4.6 turns one certificate per shape into an all-degree exclusion but does not supply it. For the grid points with a proper-power top, $(u,v) = (2,2)$, the theorem makes no claim: the odd resonances do not reduce mod $\mathbb{C}[P]$, and the certificates kill them in every tested instance (e.g. $(xy)^5$ at window 9) without a derivation of that kill.*

**Theorem 4.6** ([MV/D]; the universal tower (EXP-049)). *Let $P = x + R$ ($R$ without constant or linear part) whose total-degree top form $T = P_D$ is* not *a proper power. If the completion window is inconsistent with a certificate at any single $N \geq \deg P - 1$, then $P$ is not a Keller component at* any *partner degree. (The tower argument with $T$ the full top form: $\ker J(T,\cdot)$ on degree-$M$ forms is zero unless $\deg T \mid M$ and then the span of $T^{M/\deg T}$; and $T = P -$ lower by definition, so $T^k - P^k$ has degree $\leq k \deg P - 1$: every resonance reduces in-window modulo $\ker L$.) One cleared window solve, seconds of computation, yields an all-degree exclusion per shape; the shapes excluded in this way include multi-edge staircases such as $x+ax^2y+bx^2+cx^3y^2$ (pairing $-4620\,c^4$). For proper-power tops the measured mechanism is the* half-plane *certificate (EXP-048): a pairing-nonzero certificate supported in the $y$-heavy half-plane, disjoint from every $x$-heavy resonance source; its general construction is the remaining structural gap, and it is exactly what the $B = 16$ frontier shapes (proper-power tops) require. Certificates also exist at the open pair's degrees themselves: sampled degree-72 shapes with the GGHV corner $(16,56)$ have empty filtered windows at $N = 108$ in about one second each (EXP-050).*

**Theorem 4.7** ([MV/D]; the half-plane tower (EXP-051)). *Let $P = x + R$ whose total-degree top form attains $\min\{p-q\}$ over the support (the swallowed-staircase geometry), and let $H$ be the $y$-heavy half-plane of rows $\{i \leq j\}$. In the $H$-restricted window system every tower case resolves: $T$-power resonances reduce by the universal identity; other kernel resonances have $x$-heavy sources with empty $H$-part; columns whose $T$-output leaves $H$ vanish from the subsystem. Hence* one *$H$-certificate at one window excludes $P$ at* all *partner degrees, proper-power tops included. At the frontier, the degree-32 $B = 16$-flavor sample and the degree-72 corner-$(16,56)$ sample carry $H$-certificates (pairings $-256$, $-32$) and are excluded at all partner degrees.*

Two instrument-level results complete the toolkit. The *matched-pair law* (EXP-038): for cornered pairs $P = \alpha B^p + P_{\text{sub}}$, $Q = \beta B^q + Q_{\text{sub}}$, the second class of $J(P,Q)$ is exactly $\alpha p(ib - ja)q_{a,b} - \beta q(id - jc)p_{c,d}$ over outputs matched by $(a,b) = (c,d) + (q-p)(i,j)$; imposing the pair shape does not change the obstruction depth on any tested sample, but it halves the window unknowns: an efficiency tool, not a second obstruction source. And the *$x^m$-anchored edge operator* (EXP-043): for $E = x^m\varphi(z)$, $z = x^k y^l$,

$$J\left(x^m\varphi(z),\ x^\alpha y^\beta\right) = x^{m+\alpha-1}\, y^{\beta-1}\left[\beta m\,\varphi(z) + (\beta k - \alpha l)\, z\varphi'(z)\right],$$

verified identically in symbolic edge coefficients; Theorem 3.3's operator is the case $m = 1$.

## 5 The frontier, recalibrated from primary sources

The statements about the literature in this section were checked against the cited sources as of 2026-07-22.

*Verified gcd coverage.* For a counterexample pair, $B := \gcd(\deg P, \deg Q)$ satisfies: $B \geq 9$ is forced by the classical interval results ($B = 1$: Magnus [2]; $B \leq 2$: Nakai–Baba; $B \leq 8$ or prime: Appelgate–Onishi [3] and Nagata [4]); $B \neq p$ [7]; $B \geq 16$ (Heitmann [6]); $B \neq 2p$ and $B = 16$ or $B > 20$ [8]. Zoladek's claimed $B > 33$ has a documented gap [8] and is not used. Consequently our machine certificates at $\gcd = 2, 9, 12, 18$ (EXP-024..028) are *replications* of literature-covered territory: independent verifications, not new frontier results.

*Verified degree floor.* Moh's classical analysis discards $\max\deg \leq 100$ [5], with complete published detail only for the largest case (64); the four exceptional cases correspond to six configurations in the modern corner terminology, which the algorithmic re-examination by Guccione–Guccione–Horruitiner–Valqui enumerates and discards [10, 11]. The current floor is: any counterexample has $\max\deg \geq 125$, or $(\deg P, \deg Q) \in \{(72,108),(108,72)\}$ [11]; that single pair was left open explicitly for compute reasons.

*The open targets.* (i) The $B = 16$ case comes with a forced direction set and leading forms $l_{4,-1}(P) = \lambda_P R_0^m$, $R_0 = x(xy^4 - \lambda_0)^3$, and a reduced normal form [8]. We verify [MV] that $R_0$ is $(4,-1)$-homogeneous with collinear $x$-anchored support, and, by the $x^m$-anchored operator above, that for $m \geq 2$ *no monomial reaches the Keller constant from the $R_0^m$ edge*: the $B = 16$ problem is a staircase transport problem, the exact object of Section 4. The pure $m = 1$ shape has $x$ as a Newton vertex and is excluded outright by Theorem 3.4 (window replication empty); swallowed variants land in Theorem 4.3's territory (sampled windows empty). (ii) The similarity filter (partner supported in the scaled polygon $\frac{\deg Q}{\deg P}N^0(P)$; sound by the similarity theorem since the scaled polygons are nested in the degree) cuts the $(48,64)$ full-window system from 2142 to 111 unknowns: three degree-48 samples of the $F_1$ flavor (the pure shape $x + R_0(1)^3$ and two swallowed variants) have *empty* filtered windows at $N = 64$ in 2.4 to 3.6 seconds each, excluding all partners of degree $\leq 64$ per sample (the classical case-64 degrees: a sampled replication of Moh and Heitmann that also measures the computational cost at this scale). (iii) For the $(72,108)$ system the histograms give 5992 unknowns in 535 classes with largest diagonal block 22 before filtering: with the filter and the transport, the open territory ($\max$ degree $> 150$, and $(72,108)$ itself) is affordable at seconds-to-minutes per certificate.

### The $(72, 108)$ attack: the reduced system and the perturbative covector

The lone surviving degree pair below 125 reduces (Guccione–Guccione–Horruitiner–Valqui [11], Prop. 4.3; transcription verified against the LaTeX sources) to the existence of a pair with $[P, Q] = x^2$ supported in *small* polygons:

$$\begin{aligned} N(P):&\ (0,0),(1,0),(8,14),(8,16),(0,8) && \text{(61 lattice points)},\\ N(Q):&\ (0,0),(2,1),(12,21),(12,24),(0,12) && \text{(125 lattice points)}. \end{aligned}$$

The original paper's unsolved system was never printed. Computations EXP-050 through EXP-055 give the following. Sampled shapes at the original degrees and on the reduced polygons are all partner-free (seconds per certificate). On the stratum with the forced top edge $y^8(xy-t)^8$ symbolic, every cleared certificate pairs to the *constant* 576 (all $t$, identical across sampled lower strata; five-row support), but that covector is not rigidly universal: 51 lower combinations enter (EXP-054). This leads to the perturbative construction $\Lambda(\varepsilon)$; its base is sound (the right-hand side is the constant $x^2$ vector; $\Lambda_0$ globally left-null, symbolic $t$) and *first-order universality* is proved: all 26 entering lower monomials admit correctors with the 576 pairing pinned and localized supports. Order-1 termination fails (measured), and this pinned ladder does not terminate at any order: its descending chain stabilizes at a nonzero space of dimension 39 (EXP-064), so this route yields no universal covector; a different gauge of the pinning may terminate.

### The three reduced branches of the $(72, 108)$ case

All three branches of the GGHV reduction are certificate-covered on their forced families. Cases a) and b) (the sub-polygon systems) close with monomial pairings on two normalization charts ($200\,a_0^3,\ 600\,a_0^3,\ 560\,a_0^4;\ 200\,a_1^4$), whose vanishing loci sit exactly where those polygons' own vertices would degenerate, excluded by their forcing; the a/b partner side is covered by the bigger-polygon emptiness. Case c) closes under the torus gauge (the forced-edge parameter normalizes to $t = 1$; verified concretely by transporting an orbit sample with identical outcome): the $\beta$-symbolic certificates give pairing $23592960\,\beta$ with the degenerate $\beta = 0$ corner covered by the stratum certificates, and the interior coefficients certify axis-symbolically with constant pairings (368640 at every tested point), on top of dense mixed-direction sampling. The orientation swap is absorbed by $Q \to -Q$. Since the GGHV reduction forces *no* interior coefficients, a closure must hold for *all* interior values, and axis-symbolic coverage together with sampling does not establish that: extrapolation from slices fails in this system (the covector with constant pairing 576 on sampled strata is not universal, EXP-054). The certified statement is therefore: **the three reduced branches are certificate-covered on their forced-edge families, with interior generality axis-symbolic and densely sampled: strong evidence toward, but not a proof of, the $(72, 108)$ closure**; raising the floor to 125 requires a certificate that is simultaneously symbolic in all interior coefficients.

### The truncation ladder for the simultaneous certificate

The certified statement above reduces raising the floor to one object: a covector $\Lambda(\varepsilon)$, polynomial in the interior coefficients $\varepsilon$, that closes the reduced system for *all* interior values at once. The computations below decide this object degree by degree; every step is exact, and every modular reduction of a rational uses $\text{num} \cdot \text{den}^{-1} \bmod p$.

*Solvability is automatic (the dual annihilation identity, EXP-068).* The reduced system $M_0$ has corank one; its cokernel covector $\Lambda_0$ satisfies $\Lambda_0 M_0 = 0$ and, more, it annihilates every

Jacobian-bracket image that arises in the corrector recursion. Hence the consistency (solvability) conditions of the ladder hold at every order, and the only obstruction to a finite covector is the top-order *vanishing* condition. This is the dual form of the annihilation lemma $J(m, P^k) = -k\, L(P^{k-1}m)$ of Section 4.

*The tower, decided through degree three.* A polynomial covector of $\varepsilon$-degree $d$ exists iff a finite linear system over the interior coefficients is feasible. We find [MV]: **no covector of degree** 1 (EXP-067; eight "diagonal" operators obstruct, the full system infeasible modulo two independent primes; over $\mathbb{Q}$ this excludes every covector whose denominators are prime to both, and no good-prime argument is given for the rest); and **no covector of degree** 2 (EXP-072; the support $\{(0,1),(1,0),(3,5)\}$ obstructs, again conclusive modulo two primes). At degree 3 no obstruction is found through the pair and all triple supports (EXP-071, EXP-073), but the quadruple support $\{(0,3),(1,0),(3,4),(4,7)\}$ makes the degree-three necessary subsystem infeasible modulo both primes, the conclusive direction of a support test with the same modular caveat: **no covector of degree** 3 (EXP-075, [MV] modulo two primes). The obstruction *persists* rather than migrating: the degree-two obstructing support $\{(0,1),(1,0),(3,5)\}$ contains two of the eight operators that already obstruct at degree one, and what grows is the support size needed to expose it (single operators at degree one; at degree two every pair support is feasible, EXP-071, and a triple obstructs).

*One structural reduction, and no degree bound.* The rational-covector relaxation adds nothing (EXP-074): since $\Lambda_0 M_0 = 0$ the certificate condition is homogeneous, so any rational covector clears its denominator to a polynomial one. The dimension of the Krylov closure of $\Lambda_0$ is *not* known to bound the degree of a covector (EXP-078): the derivation of such a finite ceiling presupposes that the pinned corrector ladder terminates, and the ladder does *not* terminate (EXP-064: the descending chain stabilizes nonzero at dimension 39). Since the pinning carries a large gauge freedom, neither the ceiling nor its negation follows; *no* a priori degree bound is known, effective or otherwise, the chain of solvable degrees is controlled only by Noetherianity, which is not effective, and the degree-by-degree search is not known to be a finite decision procedure.

What does hold is weaker but exact. The set of degrees at which a covector exists is *upward closed* (a degree-$d$ covector is a degree-$(d+1)$ covector with vanishing top coefficients), so either none exists at any degree, or there is a minimal $d_0$; with degrees 1, 2 and 3 closed, $d_0 \geq 4$. And the support-restricted sweeps are *one-sided*: they test necessary conditions, so an infeasible support decides a degree negatively, while all supports passing does *not* establish existence. Settling a degree $d \geq 4$ positively therefore requires constructing a covector, not sweeping. Both outcomes remain open: a covector would be an exclusion proof, and a proof that none exists at any degree would be the first step of a counterexample.

*A lead toward a degree-uniform statement.* Written as a connection, the annihilation lemma makes the covector obstruction a first cohomology class, and termination of the corrector ladder becomes the regularity type of that connection (regular singular: the ladder terminates and a structural degree bound follows; irregular: no such bound, matching the measured non-termination of the pinned ladder). Whether this decides $(72, 108)$ in one stroke is open; we state it as a question, not a result.

## Necessary conditions from the original pair (EXP-095/096)

The reduced Proposition 4.3 system is a Laurent pair with bracket $x^2$, so theorems requiring a polynomial Keller pair with bracket one cannot be applied to it directly. Returning instead to

the hypothetical original pair gives an exact applicability bridge [MV]. Its degree-72 component has

$$2A_0 = (16, 56), \qquad 2A_0' = (2, 0), \qquad 2A_1 = (11/2, 14),$$

which is precisely the first retained $D = 72$ branch in the Newton-resolution list of Makar-Limanov–Trakhtenberg [16]. Thus that independent classification retains the GGHV branch; it does not exclude it.

Two sharper necessary conditions do result. First, instantiating Lee–Li's inner-polynomial region [14] with

$$(a, b, m, n) = (2, 3, 16, 56), \qquad \mathfrak{m} = \frac{139}{39},$$

gives [MV] exactly seven possible northeastern vertices whenever the inner or innermost polynomial is nonzero:

$$(1, 3), (2, 7), (3, 10), (4, 14), (5, 17), (6, 21), (7, 24).$$

The zero-inner-polynomial alternative remains. Nor does the stronger diagonal corollary apply, since

$$\frac{b}{a} = \frac{3}{2} \not> \frac{n - m}{a} - 1 = 19.$$

Second, the approximate-root formulas of Guccione–Guccione–Horruitiner–Valqui [13] become an exact intersection invariant [MV]. For the complete chain

$$(m, n) = (3, 2), \qquad A_1 = (11/4, 7), \qquad k = 1, \qquad l = 4,$$

the forced fourth-power edge supplies four final major classes. Each has $|D_\tau^P| = 3 \cdot 7 = 21$ and $\lambda_\tau^Q = 1/4$, hence

$$I(P, Q) = \deg_x \operatorname{Res}_y(P, Q) = 4 \cdot 21 \cdot \frac{1}{4} = 21.$$

The degree-108 component therefore has 84 major and 24 minor roots. These are necessary conditions on the original pair, independent of sampling the 51 interior coefficients of the reduced system. They do not prove that any of the seven vertices is realizable, transport either invariant through the Laurent normalization, close $(72, 108)$, or raise the degree floor. In particular, the approximate-root paper proves only an inequality for the minor-root expression where equality would be needed for the proposed family exclusion; that route is not used here.

*Transport to the reduced system (EXP-097).* These original-pair invariants cannot be imposed directly on the 51 final Laurent coefficients without additional boundary data. If $R(x) = \operatorname{Res}_y(F, G)$ and $\phi_c(x, y) = (x^{-1}, x^c y)$, then [D; exact symbolic controls]

$$\operatorname{Res}_y(\phi_c F, \phi_c G) = x^{cpq} R(x^{-1}),$$

where $p = \deg_y F$ and $q = \deg_y G$. For the final polygons $(p, q, c) = (16, 24, 4)$, so $cpq = 1536$: an input exponent interval $[\nu, 21]$ becomes $[1515, 1536 - \nu]$, of width $21 - \nu$. The reduction first swaps the selected coordinate eliminant and then localizes $x$, making $x^\nu$ a unit; Proposition 4.3 does not retain the missing $x = 0$ divisor order $\nu$. Thus absolute degree 21 remains a necessary condition for reconstructing an original polynomial pair, not a direct equation on the reduced coefficients. Laurent exponent width is a conditional replacement only after proving $\nu = 0$ and swap compatibility. A complete account of the boundary divisors through these steps could supply that missing theorem; none is claimed here.

### Constructible certificate strata and the first exact chart transition

Finite chart covers by localized certificates need a commutative-algebra qualification (EXP-098). Suppose principal opens $D(s_i)$ cover the parameter space and each carries a localized lift $c_i$ with

$$c_i^T M = 0, \qquad c_i^T b = 1.$$

Clearing denominators gives global syzygies pairing to powers $s_i^{N_i}$. Those powers generate the unit ideal, so a polynomial combination is one global covector pairing to 1. A pure principal-open cover of lifted syzygies is therefore not stronger than the global-covector target.

The strict generalization is a *constructible rank stratification*: after certifying a generic open, quotient by the residual closed-locus ideal and recompute the kernel. New syzygies can appear after this non-flat specialization and need not lift. EXP-098 verifies the distinction on an exact universally inconsistent control whose global pairing ideal is the proper ideal $(x)$, while a new certificate appears on $V(x)$.

This mechanism applies to the actual GGHV matrix (EXP-099–101). Common-flag shortcuts fail, but they isolate the first nontrivial interior interaction:

$$s = \varepsilon_{(0,1)}, \qquad t = \varepsilon_{(1,7)}.$$

For the selected $125 \times 125$ augmented minor, a Sylvester reduction gives the exact factorization [MV]

$$\frac{\det A(s,t)}{\det A_0} = \frac{(st-8)^6\big(2^{15}s^9 - (st-8)^7\big)}{2^{39}}.$$

The mixed coefficient also follows independently from

$$\operatorname{tr} B_s = \operatorname{tr} B_t = 0, \qquad \operatorname{tr}(B_s B_t) = \frac{13}{8}.$$

At $(s,t) = (-8,-1)$, the first minor vanishes, but exact ranks remain

$$\operatorname{rank} M = 124, \qquad \operatorname{rank}[M \mid b] = 125.$$

Pivot extraction gives an alternative right-hand-side-containing minor nonzero there. The first two minors cover the component $st = 8$ and leave only

$$2^{15}s^9 = (st-8)^7,$$

with normalization

$$s = 8u^7, \qquad t = \frac{8u^9+1}{u^7}, \qquad u \neq 0.$$

At $u = 1$, both previous minors vanish but a third exact augmented minor is nonzero (EXP-102). Its dense pullback has combined direction rank 121 and exceeded a five-minute time limit. EXP-103 instead evaluates at roots of unity and interpolates the complete maximal-minor polynomials by NTT. An endpoint check detects an 81-degree cancellation that a modular gcd alone would miss.

EXP-104 settles this cancellation over characteristic zero [MV]. The third determinant is evaluated exactly at 100 integer values. Its assignment bounds are $[1547, 1646]$; after division by $u^{1547}$, exact degree-99 interpolation shows that coefficients $0, \ldots, 80$ vanish and coefficient 81 does not. Thus its exact support is $[1628, 1646]$. A second row chart has exact support $[777, 903]$.

The support graphs of both charts admit connected row/column weights modulo 9 (EXP-105). With $z = u^9$, removal of the proved Laurent monomials and integer contents gives the primitive determinants

$$F(z) = 21 - 96z - 1024z^2, \qquad G(z) = (8z + 1)^{14}.$$

The only zero of $G$ is $z = -1/8$, where $F(-1/8) = 17$. More strongly, EXP-105 exhibits integer polynomials $A, B$ satisfying

$$A(z)F(z) + B(z)G(z) = 17^{14}.$$

Therefore the two normalized maximal minors generate the unit ideal over $\mathbb{Q}[u, u^{-1}]$. Together with the first two charts, this proves rank 125 for the augmented matrix at every point of the $\{(0, 1), (1, 7)\}$ coefficient slice [MV].

The grading is not confined to that slice. For every nonconstant lower monomial $x^p y^q$ tested in the 26-parameter family, EXP-106 finds the exact variable-weight law

$$w_{p,q} \equiv q - p + 1 \pmod 9.$$

All 23 remaining nonconstant directions preserve both connected chart gradings; the constant direction is bracket-zero. This supplies a coordinate system that scales to further directions, not a full-family proof. The next direction, $(0, 7)$, has selected ranks 53 and 41; after setting $z = u^9$ and $y = v/u$, both determinant charts have $z$-width 14.

EXP-107 reconstructs both bivariate minors on independent 16-by-64 NTT grids and verifies them by off-grid determinant evaluations [MV]. The endpoint-safe minor is independent of $y$, so it remains $G(z) = (8z + 1)^{14}$. The other minor restricts at $z = -1/8$ to a squarefree degree-12 polynomial $Q(y)$. Thus the first two charts leave only twelve geometric points in that fiber, not a curve.

EXP-108 selects a third 125-row chart at the residual fiber [MV]. Removing the connected mod-9 row/column weights turns both fiber matrices into exact rational matrices. Interpolation under structural degree bounds 14 and 13, using 29 exact determinants and two independent controls, gives primitive polynomials $Q, H \in \mathbb{Z}[y]$. The polynomial $Q$ is irreducible over $\mathbb{Q}$, while, up to sign,

$$\begin{aligned} H(y) = y(49y^2 + 168y + 192)(2401y^4 + 1568y^2 + 1024) \\ \cdot (117649y^6 - 403368y^5 + 921984y^4 - 1053696y^3 + 786432). \end{aligned}$$

Their exact gcd is one. The explicit integer Bézout identity has nonzero constant

$$303627789524487856472770343437884885512193900741932869903974 4.$$

Since every common zero with $G$ must lie over $z = -1/8$, the three minors prove rank 125 throughout the $\{(0, 1), (1, 7), (0, 7)\}$ coefficient slice.

These results establish the first complete constructible chart cover on an actual three-coefficient GGHV slice and expose a global lower-family grading. They do not cover simultaneous variation of the other coefficients, exclude the full $(72, 108)$ family, or raise the degree floor.

### Full-family rank and the exact 36-column core

For the full family, full rank of the coefficient matrix is an unattainable target. The $Q$-polygon contains the constant monomial, whose column is identically zero:

$$[P, 1] = 0.$$

Hence $\operatorname{rank} M(\varepsilon) \leq 124$ for every parameter value. The exact nonzero pinned augmented minor from EXP-059 then proves that $\operatorname{rank} M = 124$ and $\operatorname{rank}[M \mid b] = 125$ generically [D]. This is inconsistency on a nonempty Zariski-open subset, not on every fiber. The unresolved object is the common zero locus of augmented maximal minors. For the row pool (EXP-111), the forced-only construction has 289 rows, while the complete union inside the same reduced pool has 302 rows [MV].

EXP-112 removes the structural constant column, retains the target column, and normalizes all 51 exact perturbation matrices on a pinned 125-row chart [MV]. The union dependency graph has one 36-vertex cyclic component, three cyclic singletons, and 86 acyclic singletons. Consequently the selected determinant factors as

$$\det A_{\mathrm{sel}}(\varepsilon) = \det(A_0)(1 + \varepsilon_{(1,0)})^3 \det C_{36}(\varepsilon),$$

where $C_{36}$ depends on only 24 parameters. The other 27 directions are determinant-inert on this chart. On the forced axis,

$$\chi_{N_{(1,0)}|C_{36}}(\lambda) = \lambda^{23}(\lambda - 1)^{13},$$

recovering the total factor $(1 + \varepsilon_{(1,0)})^{16}$.

Graph support is not determinant support. EXP-113 finds two deletion-minimal triples that each make the 36-core strongly connected:

$$T_A = \{(0,1), (0,7), (2,9)\}, \qquad T_B = \{(0,1), (0,5), (1,0)\}.$$

EXP-114 computes their exact determinants [MV]. On $T_A$, the graph-active $(2,9)$ direction cancels identically, and the degree-21 determinant has only 24 monomials and factors in degrees $3, 12, 6$.

On $T_B$, write

$$a = \varepsilon_{(0,1)}, \qquad b = \varepsilon_{(0,5)}, \qquad d = 1 + \varepsilon_{(1,0)}.$$

Then

$$\det C_{36}|_{T_B} = 2^{-42} G_{54}(a, b, d) H_{63}(a, b, d),$$

where

$$\begin{aligned} G_{54} = {} & 30720000a^6b^4 + 48828125a^3b^{11} + 150000000a^3b^8d \\ & + 64000000a^3b^5d^2 - 39321600a^3b^2d^3 + 16777216d^6, \end{aligned}$$

and

$$\begin{aligned} H_{63} = {} & 4096a^9 + 184320a^6bd^2 - 1800000a^3b^5d^3 + 1843200a^3b^2d^4 \\ & + 1953125b^9d^4 + 3000000b^6d^5 + 1536000b^3d^6 + 262144d^7. \end{aligned}$$

These factors are weighted homogeneous for $\mathrm{wt}(a, b, d) = (7, 3, 9)$, of weighted degrees 54 and 63. They expose a weighted open/boundary split for the chart calculations below: normalize $d \neq 0$, then treat $d = 0$ with alternative minors from the complete 302-row system. No result in this subsection excludes the residual factor loci, the full $(72, 108)$ family, or $JC(2)$.

### Weighted residual transitions and the $d = 0$ quotient

EXP-115 first normalizes the open chart $d \neq 0$ by

$$a = Au^7, \qquad b = Bu^3, \qquad d = u^9.$$

Writing $X = A^3$, $G_{54}(A, B, 1)$ remains irreducible, while

$$H_{63}(A, B, 1) = L(A, B)Q(A, B)$$

splits into irreducible degree-three and degree-six factors over $\mathbb{Q}[A, B]$ [MV]. Exact good-prime alternative minors prove that none of $G_{54}, L, Q$ is wholly contained in the rank-deficient locus. The $L$-transition also lifts to the rational point $(A, B, d) = (0, -4/5, 1)$. Proper intersections on these components remain.

The boundary $d = 0$ is structural. Cancelling the forced $x$-term puts

$$P = ay + by^5 + y^8(1 - xy)^8$$

inside the retained $Q$-space, so $[P, P] = 0$ is an exact polynomial right kernel [D]. Its $y^8$-coordinate is the unit 1; deleting that column gives a 302-by-124 quotient augmented matrix with 123 coefficient columns. The origin has exact rank profile 112/113, while the nonzero controls $(0, 1)$, $(1, 0)$, and $(-9, 12/5)$ have profile 123/124 [MV].

EXP-116 compresses the quotient graph into cyclic blocks of sizes

$$51, 11, 10, 9, 8, 7$$

and sixteen cyclic singletons. EXP-117 computes the determinants of the 51-block and of all smaller blocks. In original coordinates the selected quotient determinant is, up to an explicit nonzero rational scalar,

$$\det A_{\text{quot}} \doteq b^{32} F_{28}(a, b) \prod_{c \in \mathcal{C}} (ca^3 + 78125b^7), \qquad \mathcal{C} = \{49152, 32768, 16384, 8192, 4096\},$$

where

$$\begin{aligned} F_{28} = {} & 1375061556226896691 2a^{12} - 3892296469079654400000000a^9b^7 \\ & + 6149594411200000000000000a^6b^{14} \\ & + 8653485107421875000000000a^3b^{21} \\ & - 7119029760360717773437 5b^{28}. \end{aligned}$$

All factors are homogeneous for $\text{wt}(a, b) = (7, 3)$. On $b \neq 0$, set $z = a^3/b^7$. The residual becomes one squarefree degree-nine equation: five rational roots

$$-\frac{78125}{4096}, \quad -\frac{78125}{8192}, \quad -\frac{78125}{16384}, \quad -\frac{78125}{32768}, \quad -\frac{78125}{49152},$$

and four roots of the quartic obtained from $F_{28}/b^{28}$ [MV]. Thus the selected quotient residual is reduced to $b = 0$ and nine invariant points.

EXP-118 supplies the exact alternative chart [MV]. Its complete quotient determinant is

$$\det A_{\text{alt}} \doteq a^{107}.$$

The scalar is a nonzero integer. This chart covers $a \neq 0$, including all nine invariant points. On $a = 0$, the selected determinant is exactly $b^{95}$, and at the origin the quotient rank profile is 112/113. These three cases close the complete $d = 0$ quotient plane. This is a boundary-stratum closure, not a result about $d \neq 0$.

### Exact weighted-open charts and component ideals

EXP-119 verifies the full 302-by-125 augmented matrix is covariant for $\mathrm{wt}(a,b,d) = (7,3,9)$ [MV]. On $d = 1$, a first alternative row basis gives an exact determinant with 114 monomials. Writing $X = A^3$, it factors as a nonzero rational scalar times

$$B^{36}K(X,B) \prod_{c\in\{4096,8192,16384,32768,49152,65536,86016\}} (cX + 78125B^7),$$

where $K$ has degree 6 in $X$, degree 22 in $B$, and 30 monomials. Its gcd with each irreducible selected component $G, L, Q$ is one. Nevertheless, the three exact elimination resultants are nonconstant, of degrees 852, 324, and 648 in $B$. Hence this chart removes every curve component but leaves three finite proper intersections.

EXP-120 evaluates the independently selected $G$-basis [MV]. The first rational normalization $(A,B,d) = (1,0,1)$ works, the largest cyclic block has size 25, and the exact determinant has 21 monomials. Up to a nonzero rational scalar its invariant form is

$$\Delta_G = X^{30}L(X,B)Q(X,B)R(X,B),$$

where

$$R = -156250B^7 + 15625B^6X + 28000B^3X + 3200B^2X^2 + 1024X.$$

Factorwise exact Gröbner calculations give

$$(G,\Delta_{LQ},L) = (G,\Delta_{LQ},X) = (G,\Delta_{LQ},Q) = (G,\Delta_{LQ},R) = (1).$$

Therefore the three charts cover the complete $G$ component [D].

The corresponding EXP-120 ideals on $L$ and $Q$ are nonunit but zero-dimensional. The $L$ lexicographic eliminant has degree 108 in $B$ and squarefree degree 73. For $Q$, a six-element graded-lex basis proves zero-dimensionality; an FGLM conversion exceeded a 180-second time limit, so no lex eliminant or point count is claimed. The explicit $LQ$ factor in $\Delta_G$ shows that this basis cannot remove either remaining component. The next exact charts must be selected directly on those finite residual schemes.

EXP-121 performs that selection [MV]. At $p = 1013$, thirteen affine points of the $L$ residual have coefficient/augmented rank profile 124/125; at $p = 1033$, eighteen points of the $Q$ residual have the same profile. Deterministic row extraction on the two components returns the same 125-row basis, although it differs from the pinned basis by 68 rows. Its rational anchor is $(A,B,d) = (1,0,1)$, its largest cyclic block has size 26, and five direct 125-by-125 controls agree with the block product. The exact determinant has total degree 108 and 23 monomials:

$$\Delta_{\mathrm{fin}}(A,B) = A^{87}R_{\mathrm{fin}}(A^3,B), \qquad \deg_X R_{\mathrm{fin}} = 5, \quad \deg_B R_{\mathrm{fin}} = 18.$$

For $L$, the exact graded-reverse-lexicographic calculation returns

$$(L,\Delta_{LQ},\Delta_{\mathrm{fin}}) = (1).$$

For $Q$, a direct Gröbner calculation on the three generators exceeded a 240-second time limit, so the certificate uses the exact factor $\Delta_{LQ} = B^{36}\Delta^\circ$. On $B = 0$,

$$\gcd\big(256X^2 - 1024X + 4096,\ X^{32}\big) = 1.$$

On $\Delta^\circ = 0$, reduction modulo the quadratic $Q$ makes both $\Delta^\circ$ and $\Delta_{\mathrm{fin}}$ linear in $X$. The exceptional coefficient gcd is one, and the resulting degree-144 and degree-176 compatibility polynomials in $\mathbb{Q}[B]$ also have gcd one. Consequently

$$(Q, \Delta_{LQ}, \Delta_{\mathrm{fin}}) = (1).$$

This is an exact zero-set split and univariate gcd certificate, not a modular extrapolation.

Together, EXP-118–121 close the complete three-parameter $T_B$ restriction: EXP-118 covers $d = 0$, EXP-120 covers $G$ on $d \neq 0$, and EXP-121 covers the remaining finite $L/Q$ strata. They do not close the 24-parameter core, the full 51-parameter family, $(72, 108)$, the planar degree floor, or $JC(2)$.

## First higher-dimensional lift from the cyclic core

EXP-122 restores the other 21 directions in the 24-parameter cyclic core on the shared EXP-121 basis [MV]. No restored direction is determinant-inert on its complete anchor line. Thirteen directions act inside the existing size-26 cyclic block, 16 enlarge the fixed-$d$ SCC, and every restored direction contributes at first order or pairwise mixed order. Hence, on this basis, no restored direction lifts for free.

The eight directions $(2, j)$, $2 \leq j \leq 9$, nevertheless form a sparse class: each enlarges the SCC but has a linear anchor-line determinant. Direction $(2, 9)$ has the smallest selected rank, normalized support, and enlarged SCC. EXP-123 computes its complete symbolic lift on $d = 1$ [MV]. Writing $C = \varepsilon_{(2,9)}$,

$$X = A^3, \qquad Y = A^2 C,$$

the shared determinant is

$$\Delta_{(2,9)}(A, B, C) = A^{87}\big(R(X, B) + Y S(X, B)\big).$$

Here $R$ is the 23-term EXP-121 polynomial, $S$ has 18 terms, and

$$\gcd_{\mathbb{Q}[X,B]}(R, S) = 1.$$

Two modular probes precede the exact computation. The claim rests on the characteristic-zero determinant, the check that its specialization at $C = 0$ reproduces the complete EXP-121 determinant, and four direct full-matrix rational controls.

Consequently, on $A \neq 0$ and $S \neq 0$, the selected exceptional locus is the rational graph

$$Y = -\frac{R(X, B)}{S(X, B)}.$$

The base locus $V(R, S) \subset \mathbb{A}^2_{X,B}$ is zero-dimensional because $R$ and $S$ are coprime. This is the first higher-dimensional constructible reduction from the 24-parameter core [D]. By itself it is not a four-parameter chart cover: alternative minors must cover the rational graph, the finite base locus, and the separate divisor $A = 0$. The graph and the base locus are closed below. The complete $(72, 108)$ case, the planar degree floor, and $JC(2)$ remain open.

### Recursive atlas on the rational graph

EXP-124 selects an alternative complete-row basis directly on the graph [MV]. The same one-row replacement is full rank over two good primes, and its exact determinant is

$$\Delta_{\mathrm{alt}}(A,B,C) = A^{90}N(A^3,B).$$

It is independent of $C$. The 21-term polynomial $N(X,B)$, of total degree 16, factors over $\mathbb{Q}$ as a nonzero scalar times

$$N(X,B) = F_3(X,B)F_6(X,B)F_7(X,B),$$

where the factors have total degrees $3,6,7$ and are each coprime to $R,S$. Thus this minor covers the dense graph open $N \neq 0$, reducing the positive-dimensional residual to three explicit plane curves.

EXP-125 recurses first on

$$F_3(X,B) = (5B+4)^3 + 16X.$$

Suitable primes are not automatic: cubic-residue and factor-point obstructions exclude some primes, and sample points must lie on the cube locus because $X = A^3$ is required. The primes 739 and 811 satisfy these conditions. On both primes, four points on each of $F_3, F_6, F_7$ have coefficient/augmented rank 124/125, and each curve supplies a new cross-prime basis [MV].

The exact $F_3$-selected determinant has one cyclic block of size 32. After restriction to $Y = -R/S$ and substitution $X = -(5B+4)^3/16$, its univariate numerator factors as

$$U_3(B) \doteq (5B+4)Q_6(B)^2Q_9(B)Q_{15}(B).$$

Here $\doteq$ denotes equality up to a nonzero rational scalar and the subscripts are exact irreducible degrees. The linear factor forces $A = 0$; $Q_6$ divides both $R|_{F_3}$ and $S|_{F_3}$; and $Q_9, Q_{15}$ divide neither. Consequently, on $AS \neq 0$, only

$$Q_9(B)Q_{15}(B) = 0$$

remains on $F_3$: 24 normalized parameter values and 72 lifted points over an algebraic closure [MV/D]. This is a finite reduction of one component, not a complete four-parameter cover.

EXP-126 treats a maximal minor as a section on the next residual curve [MV]. The EXP-125 $F_6$ basis is exactly the $F_3$-selected basis whose characteristic-zero determinant was already reconstructed. After hash, anchor, SCC, graph-numerator, modular-sample, and direct-determinant checks, the graph numerator has the nonzero class

$$h_6(X,B) = U(B)X + V(B) \quad \text{in} \quad \mathbb{Q}[X,B]/(F_6).$$

The function-field norm is computed independently by a Sylvester resultant and by the determinant of multiplication by $h_6$ in the quadratic quotient. It has degree 74 and factors, up to scalar, as

$$\mathrm{Norm}(h_6) \doteq D_2D_3^4D_6^2Q_{18}Q_{30}.$$

The five factors are irreducible, with subscripts denoting degrees. Direct arithmetic in the corresponding quotient fields shows that $D_2$ gives $X = S = 0$, while $D_3, D_6$ give $R = S = 0$. The degree-3 case is a double $F_6$ fibre on which $U = V = 0$, so it is classified at the double

root rather than by a projection resultant alone. Neither $Q_{18}$ nor $Q_{30}$ meets $X = 0$ or $S = 0$. Consequently, on $AS \neq 0$, the effective residual is

$$Q_{18}(B)Q_{30}(B) = 0.$$

This gives 48 normalized values and 144 lifted algebraic points [MV/D]. It is the first explicit use of the maximal-minor / curve-section / zero-divisor / function-field-norm dictionary in this atlas.

EXP-127 applies the same dictionary to the distinct EXP-125 $F_7$ basis [MV]. Its exact anchor has a 31-vertex largest cyclic block. The complete determinant has $A$-valuation 89 and, after invariant reduction, is independent of $Y = A^2C$. Modulo the irreducible quadratic-in-$X$ curve $F_7$, its primitive class is again linear:

$$h_7(X, B) = U_7(B)X + V_7(B).$$

The Sylvester resultant and quotient-multiplication determinant give the same degree-58 norm, whose irreducible factorization is

$$\mathrm{Norm}(h_7) \doteq B^{16}E_3E_9E_{12}E_{18}.$$

Direct quotient arithmetic identifies the $B = 0$ common root as $X = 0$, and the $E_{12}$ root as a point of $R = S = 0$. The unique roots over $E_3, E_9, E_{18}$ have $X, S \neq 0$. Hence the effective residual on $AS \neq 0$ is

$$E_3(B)E_9(B)E_{18}(B) = 0,$$

giving 30 normalized values and 90 lifted algebraic points [MV/D]. Together with the $F_3$ and $F_6$ reductions, this finishes the positive-dimensional factor-curve phase of the graph atlas.

EXP-128 computes the finite union in its squarefree quotient algebra [MV]. The seven retained factor appearances contain five distinct projected factors of degrees

$$9, 15, 18, 30, 3,$$

and their product has degree 75, not 102. The degree-9 factor is shared by the $F_3$ and $F_7$ residuals; the degree-18 factor is shared by the $F_6$ and $F_7$ residuals. Exact CRT idempotents verify the five-factor direct product. Restricting $h_7$ to the other curves shows that it is a unit on the unique degree-15 $F_3$ block and the unique degree-30 $F_6$ block. Its only failures there are the two shared blocks. The reciprocal $h_{36}$ section vanishes identically on $F_7$, so a new section is required on the complete degree-30 $F_7$ divisor.

EXP-129 first verifies that the shared factors also have identical exact $X$-classes in their two curve descriptions [MV]. Two finite-field probes on each $F_7$ block have augmented row rank 125. No single sampled row basis covers all six probes, so one section does not suffice; a two-basis atlas does. Both bases reconstruct in characteristic zero by the same SCC/block-determinant method. If their primitive quotient sections are $k_1, k_2$, exact norm gcds and Bezout identities give

$$\mathrm{Norm}(k_2) \in \big(\mathbb{Q}[B]/(E_3)\big)^{\times}, \qquad \mathrm{Norm}(k_1) \in \big(\mathbb{Q}[B]/(E_9)\big)^{\times} \cap \big(\mathbb{Q}[B]/(E_{18})\big)^{\times}.$$

Thus their norm ideal is the unit ideal in every block of $\mathbb{Q}[B]/(E_3E_9E_{18})$. Combined with EXP-124–128, this proves that the maximal-minor atlas covers the complete $AS \neq 0$ rational graph [MV/D].

EXP-130 treats the finite base locus as a coordinate algebra rather than as a list of algebraic points [MV]. Both projection resultants have degree 117. After separating their coordinate factor of multiplicity 27, their squarefree principal-open factors have degrees

$$3, 6, 12, 69.$$

Factorwise subresultants reconstruct the same-point $X$-class on every factor, proving that

$$K = \big(\mathbb{Q}[X, B]/(R, S)\big)[X^{-1}]$$

is a reduced product of four fields of total dimension 90. Direct specialization gives

$$\gcd(R(0, B), S(0, B)) = B^{12}, \qquad \gcd(R(X, 0), S(X, 0)) = X^3.$$

This places the discarded coordinate support at $X = B = 0$, on the separate $A = 0$ boundary. The projection multiplicity is not used as a local-length claim.

The EXP-124 section is a unit on the degree-69 block, while the EXP-125-$h_{36}$ and EXP-129-atlas-1 sections generate the unit ideal on degree 12. The first newly selected exact basis leaves one common quadratic in $K[Y]$ on each of the degree-3 and degree-6 blocks. Modular probes at the roots of those quadratics have full augmented rank, but reconstruction of the default targeted basis exceeded a 300-second time limit. Starting instead from the EXP-125-$F_3$ basis and replacing one row reduces the largest determinant SCC to 33. Its characteristic-zero determinant has the form

$$A^{86} H(X, B, Y), \qquad \deg_Y H = 2,$$

and agrees with a direct 125-by-125 determinant at an independent control point [MV].

On both remaining fields, this section breaks the common quadratic. Exact extended Bezout identities in $K[Y]$ and independent $Y$-resultants certify the final pairs. Multiplication matrix norms provide an additional exact control and show that the final pairs generate the unit ideal [MV]. Hence the finite maximal-minor atlas covers $V(R, S) \cap D(X)$ uniformly in $Y = A^2C$. Combining the selected EXP-123 chart, the EXP-129 graph atlas, and the EXP-130 base-locus atlas exhausts every case with $A \neq 0$ [D].

EXP-131 returns to the original 302-by-125 augmented matrix before any division by $A$ and specializes directly at $A = 0, d = 1$ [MV]. Two exact row bases give

$$\Delta_1 = \kappa_1 (5B + 4)^3 (25B^2 - 20B + 16)^3$$

and

$$\Delta_2 = \kappa_2 B^{95} (109375B^6 - 110592)(21875B^6 - 4800B^3 - 24576),$$

with $\kappa_1, \kappa_2 \in \mathbb{Q}^\times$; both are independent of $C$. For the squarefree divisors

$$f = (5B + 4)(25B^2 - 20B + 16), \qquad h = B(109375B^6 - 110592)(21875B^6 - 4800B^3 - 24576),$$

exact extended Euclid gives

$$1 = \left( \frac{478515625}{17179869184} B^{12} - \frac{2734375}{134217728} B^9 - \frac{205625}{4194304} B^6 + \frac{1025}{32768} B^3 + \frac{1}{64} \right) f - \frac{25B^2}{17179869184} h.$$

Thus the two augmented minors generate the unit ideal in $\mathbb{Q}[B,C]$, so the complete $A=0,d=1$ plane is covered [MV/D]. EXP-118 covers $d=0$, and EXP-123/129/130 cover $A\neq 0,d=1$. Therefore the complete four-coefficient restriction $\{(0,1),(0,5),(1,0),(2,9)\}$ is closed [D]. The 24-parameter core, complete 51-parameter family, $(72,108)$ case, planar degree floor, and $JC(2)$ remain open.

EXP-132 next adjoins $T=\varepsilon_{(2,8)}$ and works on the same original 302-by-125 augmented matrix at $A=0,d=1$ [MV]. Three normalized maximal minors are

$$\begin{aligned} P(B) &= (5B+4)^3(25B^2-20B+16)^3,\\ Q(B,C,T) &= B^{95}\big(4785156250B^{12}+9765625B^{11}CT-1050000000B^9\\ &\qquad -10214400000B^6+1061683200B^3+5435817984\big),\\ R(B,C) &= B^{105}C. \end{aligned}$$

Consequently

$$Q-9765625\,BTR=Q_0(B),\qquad \gcd(P,Q_0)=1,$$

and the extended-Euclid computation gives explicit $u,v\in\mathbb{Q}[B]$ satisfying

$$uP+v\big(Q-9765625\,BTR\big)=1.$$

Thus $(P,Q,R)=\mathbb{Q}[B,C,T]$, and the complete direct $A=0,d=1$ boundary of the five-coefficient restriction $\{(0,1),(0,5),(1,0),(2,9),(2,8)\}$ is closed [MV/D]. The primary section is exactly $T$-inert; the alternative is affine in $T$. The final basis has 106 singleton cyclic components and a monomial determinant. In contrast, the inherited EXP-123 shared basis and EXP-124 alternative basis both vanish identically on this direct specialization, so neither inherited section suffices.

There is a useful module-theoretic reading: the augmented maximal minors are the zeroth Fitting ideal of the cokernel, and the certificate above proves that this specialized presentation is everywhere surjective. Recent work on multivariate polynomial matrices gives Smith-form equivalence criteria for a structured class with triangular highest determinant and extensions to rectangular or rank-deficient matrices [15]. We have not established those structural hypotheses for the GGHV augmented matrix, so that theory is a comparison lens and a possible source of local reduction lemmas, not a shortcut or an additional closure claim. The $A\neq 0,d=1$ fifth-direction atlas is partly lifted: EXP-135 retains the EXP-124 section exactly, while the two EXP-129 residual sections and EXP-130 finite-base section remain. The transverse $d=0$ kernel quotient also remains open; hence the complete five-coefficient restriction, 24-parameter core, full family, $(72,108)$, the planar degree floor, and $JC(2)$ remain open.

**Proposition 5.1** (Ambient transverse inertness of the EXP-124 section)**.** *Let*

$$H(A,B,C)=I+(A-1)K_{(0,1)}+BK_{(0,5)}+CK_{(2,9)}$$

*be the normalized 33-by-33 cyclic core of the EXP-124 section, and write $K_T=K_{(2,8)}$. Then*

$$\det\big(H(A,B,C)+TK_T\big)=\det H(A,B,C)\quad \text{in } \mathbb{Q}[A,B,C,T].$$

*The remaining 86 singleton blocks are also $T$-independent, so the complete selected normalized maximal minor is unchanged by $(2,8)$.*

*Proof.* Exact row reduction gives $\operatorname{rank} K_T = 7$, $\operatorname{rank} K_T^2 = 3$, and $K_T^3 = 0$. Fix a rank factorization $K_T = UV^{\mathsf{T}}$. The ranks of the four coefficient matrices in $(A, B, C, T)$ are respectively $(25, 24, 6, 7)$, so these are separate degree bounds for the determinant difference. At each point of the tensor grid

$$A \in \{1, \dots, 26\}, \qquad B \in \{0, \dots, 24\}, \qquad C \in \{0, \dots, 6\},$$

and at each of 30 distinct primes, exact modular elimination finds $H$ invertible and verifies

$$\left(V^{\mathsf{T}} H^{-1} U\right)^2 = 0.$$

The matrix determinant lemma therefore makes $\det(I_7 + TV^{\mathsf{T}} H^{-1} U) = 1$ as a polynomial in $T$ at every grid point. Tensor interpolation proves that every coefficient of the cleared determinant difference is zero modulo every listed prime.

For the characteristic-zero lift, the common entry denominator is $D = 5{,}241{,}600$. The product of the rowwise coefficient $\ell^1$-norms of $D(H + TK_T)$ bounds the determinant coefficient $\ell^1$-norm. Including the subtraction at $T = 0$ gives an explicit 885-bit absolute coefficient bound. The 30-prime product has 897 bits and exceeds twice that bound. A nonzero integer coefficient cannot therefore vanish modulo their product. Every coefficient is zero, proving the identity over $\mathbb{Q}$ [MV/D]. The exact singleton calculation is EXP-134. □

The proof changes the computational viewpoint. It avoids both a dense 33-by-33 symbolic determinant and graph-function-field inversion: rank gives the interpolation degrees, low-rank determinant and Woodbury lemmas control each finite evaluation, and a height bound turns modular vanishing into an exact rational identity. The same route applies to the EXP-129 section, which is observed to be inert; sections that are affine or quadratic in $T$ require their actual $T$-coefficients, since inertness is not inherited.

*Beyond degree 125.* The chain $(8, 40)/(8, 28)/(11/4, 7)$, of maximal degree 144, is already excluded in the published literature: the closing remark of [9] states that the corner data $B_0 = (8, 28)$, $B_1 = (8, 40)$ force $A_0' = (8, 4)$, which is not a possible last lower corner (EXP-082). Of the other twenty-three configurations with maximal degree in $[125, 150]$ (EXP-083), four share a corner with the cases of that remark but fail its hypotheses, so the remark does not exclude them (EXP-094). None of the twenty-three is decided by these sources; for sixteen of them the last lower corner $A_0'$ is not printed in the literature and has to be derived before membership in the excluded family $\wp(n', n' - 1)$ can be tested (EXP-083).

## The counterexample as an incidence map, and its exclusion in the plane

A complementary lens sharpens why the two-variable case resists the very mechanism that settled dimension three. The 2026 counterexample is not an arbitrary polynomial map: it is the coefficient map of the product of a linear and a quadratic binary form, restricted to a normalized slice. Concretely, on $V_1 \times V_2 \to V_3 \times \mathbb{A}^1$, $(L, Q) \mapsto (LQ, \operatorname{Res}(L, Q))$, the slice $X_H = \{(L, Q) : \operatorname{Res}(L, Q) = 1,\ [LQ]_{\text{next}} = 1\}$ carries the projection $(L, Q) \mapsto LQ$, which is étale on the coprime (nonzero-resultant) locus by a one-line argument: multiplication forgets exactly the relative scaling $(L, Q) \mapsto (\lambda L, \lambda^{-1} Q)$, and the resultant, of weight $\deg Q - \deg L$ under that scaling, detects it. The projection is generically $n$-to-one (the choices of linear factor), hence not injective. We verified [MV] the full identification: the displayed counterexample equals this coefficient map up to linear normalization, with $X_H \cong \mathbb{A}^3$. So, for this family, the entire Jacobian question reduces to a single *recognition of affine space*: is $X_H \cong \mathbb{A}^n$?

In dimension three the recognition holds by a toric coincidence: the residual $\mathbb{G}_m$-grading has weights $(1, -1, -2)$, and the associated semigroup generators form a unimodular lattice basis, so $X_3 \cong \mathbb{A}^3$. In the plane it *fails*, and we can say why in one invariant. The controlling quantity is $n - 2 = \deg Q - \deg L$, the resultant weight under the relative scaling; it is simultaneously the degree of the Danielewski core $\{x^{n-2}W = B^{n-2} - 1\}$ and the number of boundary components. For $n = 2$ it is 0: the resultant becomes scaling-invariant, unable to rigidify the normalization, and the slice degenerates (reducible; equivalently, the finite-root model is $\mathbb{A}^2$ minus the discriminant, which carries a nonconstant unit and is therefore not $\mathbb{A}^2$). For $n = 3$ it is 1, the unique value that is both nonzero and linear, giving $X_n \cong \mathbb{A}^n$; for $n \geq 4$ it is $\geq 2$, and the class group obstructs. Thus the exact structure of the known counterexample cannot occur in the plane [MV], a conclusion consistent with the $\mathbb{G}_m$-equivariant rigidity theorem of the companion paper [22], which forbids the torus-graded planar analog independently. This excludes the incidence/root-selection class, the structure of every constant-Jacobian noninvertible map known to date; it is strong structural evidence for $JC(2)$, not a proof, since a planar counterexample of some entirely different structure is not addressed.

## 6 Positioning and novelty

A search of the literature (the sources of Section 5) found no prior statement of Theorems 3.1 or 3.2. For the equivariant rigidity theorem of the companion paper [22], Shaska [21], in work submitted on 2026-07-22, independently proves the full all-signature classification. Same-sign actions may give nonlinear triangular automorphisms, one zero weight gives affine maps, and opposite-sign actions give the linear case of EXP-010. No priority or novelty claim is made for that classification. Theorem 3.3 was not found as stated, but its operator identity is within the standard bracket calculus [17, 8]: the statement appears new, the method is classical. Theorem 3.4's counterexample half is essentially contained in [8] (Prop. 4.1) with van den Essen's subrectangular normalization [18]; we claim only the sharp dichotomy form. All proofs are elementary once seen, and some of these statements may be folklore; the primary sources consulted, including the items that could not be verified, are listed in the repository.

## 7 Open problems

The computations above leave the following problems open: (1) lift the fifth direction $(2, 8)$ through the two EXP-129 residual sections and the EXP-130 finite-base section; the complete-grid CRT argument of Proposition 5.1 applies wherever explicit degree and height bounds are available; (2) construct the transverse $d = 0$ kernel quotient and determine its Fitting ideal; (3) transport the intersection number 21 to the reduced system, which requires a complete account of the boundary divisors through the swap, the Laurent cuts and the final inversion; and (4) derive a condition beyond the first $D = 72$ branch of the Newton-resolution list, which retains the GGHV branch; without such a condition, Newton resolution does not exclude the pair. The properness reformulation ($JC(2)$ holds iff every planar Keller map has empty Jelonek set; exact certificates in EXP-014) provides an independent cross-check.

## 8 The $(72, 108)$ frontier: an external exclusion claim and its independent verification

Helali [12] states that the two coefficient systems of [11, Proposition 4.3], $[P, Q] = x^2$ in $K[x, x^{-1}, y]$ with

$$N(P) = \mathrm{conv}\{(0,0), (1,0), (8,14), (8,16), (0,8)\},$$
$$N(Q) = \mathrm{conv}\{(0,0), (2,1), (12,21), (12,24), (0,12)\}$$

(Case 1) and the same polygons without $(0, 8)$ and $(0, 12)$ (Case 2), have no solution, by exact unit-ideal and membership identities over number fields, and concludes, conditionally on the exhaustiveness of that proposition and on the transcription, that a plane counterexample has maximal degree at least 125. We checked the claim with code written from the definitions, reading his serialized data only as data (EXP-137).

*Equations* [MV]. With $t = xy^2$, $z = y^{-1}$ one has $[t, z] = -1$ and $x^2 = t^2z^4$. Our band equations agree with his displayed equations for Case 2 and, for Case 1, in the three top bands; in the two lowest upper bands of Case 1 they differ only by the cross terms of the negative bands, which his reduction treats in later steps. Case 1 has 302 coefficient equations.

*First elimination* [MV]. With $A = t + a_2t^2 + \cdots + a_7t^7 + t^8$, the top band determines $D = d_2t^2 + \cdots + d_{12}t^{12}$ and leaves six compatibility equations whose ideal is zero-dimensional of degree 35; its eliminant $H(a_7)$ equals his, is irreducible over $\mathbb{Q}$, and is $G(a_7^7)$ for an irreducible quintic $G$ whose field is isomorphic to his $\mathbb{Q}[w]/(w^5 - w^4 + 3w^3 + 3w^2 + 26)$.

*Case 2* [MV]. Our layered solve over $\mathbb{Q}[a_7]/(H)$ leaves 25 equations in three parameters whose standard basis is $\{1\}$, so Case 2 has no solution; his four residual generators are among our equations.

*Case 1* [MV for the identities, evidence for emptiness]. Our 13 compatibility equations span the same space as his, and the first is $\kappa(s^2 - \gamma)^2$, so his split $s = \pm c$ is exhaustive. All of his identities for this case hold in our exact arithmetic over the degree-35 field. We did not reproduce his reduction from these 13 equations to the systems his identities concern, and our Gröbner computation over the field did not finish within the time allowed. Independently of his data, at the primes $1021, 1031, 1033, 1049, 1061$, where $H$ has a simple root, both branches of our 13 equations reduce to the unit ideal over $\mathbb{F}_p$, while the subsystem of the upper layers does not. This is strong evidence, not a proof, that Case 1 is empty.

Conditionally on [11, Proposition 4.3] being exhaustive, the exclusion of $(72, 108)$ therefore has an independent proof of its second half and independent verification and evidence for its first half. Priority for the exclusion is Helali's.

## Data and code availability

All computations are exact and reproducible from the repository
`https://github.com/fsantibanezleal/CAOS_RESEARCH`. An identifier EXP-*nnn* denotes one archived record in the experiment folders of the problem,
`problems/algebraic-geometry/jacobian-conjecture/experiments/`; each record states its question, method and result with its independent checks, together with the scripts and outputs of its computations, and the larger computations list SHA-256 hashes of their result files.
Every [MV] claim has an archived script and output.

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